Course Syllabus
ECON 4050: Introduction to Econometrics – Fall 2026

Instructor: Dr. Adam Soliman
E-mail: asolima@clemson.edu
Office: Powers Hall 320K
Department Phone: 864-656-3481

Teaching Assistants: Caleb Vines — **Email**: cmvines@clemson.edu
Haoran Li — **Email**: haoran2@clemson.edu

Lecture Schedule and Location
ECON 4050-002 Tuesday and Thursday 12:30PM-1:45PM Powers Hall G016

Lab Schedule
ECON 4051-001 Tuesday 5:30PM-8:30PM (Caleb)
ECON 4051-002 Wednesday 5:30PM-8:30PM (Haoran)

Student Hours
Dr. Soliman: On Zoom. Please sign up for a slot by clicking here.
TA: Caleb will hold them Tuesdays from 4-5pm in Powers 318G and Haoran on Fridays from 12-1pm in Powers 314B.

Course Description and Learning Objectives
ECON 4050 introduces students to the study of modern econometric techniques as employed in economics and policy analysis. Throughout the course, we will discuss how these techniques can be used to evaluate data and conduct policy analysis including the potential problems and pitfalls with doing so. The course will cover both theoretical and practical issues. Problem sets will contain applications to real data and require the use of R (a statistical software).

There are several objectives for the course:
1) Be able to understand statistical data presentations using averages, standard deviation, t-statistics, and hypothesis tests.
2) Conduct simple and multiple linear regression analyses.
3) Interpret coefficients and p-values from regression outputs.
4) Test hypotheses using confidence intervals, t-tests, and F-tests.
5) Apply econometric techniques to real-world data using software tools (R).
6) Distinguish between correlation and causation, and understand the assumptions required for causal inference.

While the material covered in ECON 4050 is typically considered challenging, the tools are highly useful. You will be able to estimate the relationship between any two variables, such as years of education and wages, exercise and blood pressure, studying and grades, campaign spending and election results, and so on. You will be able to estimate these relationships after controlling for many other confounding variables, such as intelligence, family history, gender, race, occupation, obesity, cholesterol, and so on. You will be able to evaluate the effect of government policies, such as minimum wage, retirement ages, subsidies, minimum sentencing legislation, additional police, immigration reform, access to health insurance, and so on. These tools are also readily applicable to business questions such as the effects of marketing campaigns, employee productivity, profit forecasting, training programs, and wage bonuses.

Course Materials
All relevant materials can be found on the following course website on Github:
https://adamsoliman.github.io/IntroEconometrics/
Note there is no required textbook for the course. All materials are free and available online.

Pre-requisites
ECON2110 and 2120 Principles of Microeconomics and Macroeconomics
MATH1080 Calculus One
STAT3090/MATH3090 Introduction to Statistics

Co-requisites:
ECON4051/6051 Introduction to Econometrics Laboratory Course Overview

Grading Activities and System
Grades will be based on lab assignments, assessments, midterm exams, and a final project that includes a presentation. The following weights will be applied to determine your final grade:

Lab Assignments	12.5%
Lecture-Related Assessments	12.5%
Coding Midterm	20%
Theory Midterm	20%
Final Project Presentation	15%
Final Project	20%

The TAs and I will attempt to complete grading of each assignment within one week of the due date. Afterwards, you will have one week to review the grading and appeal any errors that you think have been made. To appeal a grade, you should e-mail the person who graded the assignment and provide a detailed explanation for why you believe there is an error. *Note that any regrading may either increase, decrease, or not change your grade, and that any answer that cannot be read will receive a zero.*

Final grades will be calculated using the weighting scheme listed above, regardless of how Canvas displays the running course average. Canvas may not always display the weighted grade correctly during the semester, especially if assignment groups or point values differ. *Students are responsible for reviewing the grading weights in the syllabus and asking questions if their understanding of their current grade differs from what is displayed in Canvas.*

A	90-100%
B	80-90%
C	70-80%
D	60-70%
F	<60%

Lecture Attendance and Assessments

Regular assessments for lecture will take two forms:

- *Warm-up quizzes*: Each chapter begins with a short multiple-choice quiz reviewing the previous chapter's content. These are completed in class at the start of lecture. Missed quizzes generally cannot be made up unless the absence is excused.
- *Out-of-class tasks*: Each chapter includes applied tasks using real data in R. You are expected to complete these on your own time, using whatever tools or resources you find helpful, and submit your work to Canvas before the next class. I will randomly select students to present and explain their answers during class, so you should be prepared to walk through your approach. Consistent, thoughtful submissions and willingness to engage during presentations factor into this component of your grade.

Lecture attendance is not formally required, but quizzes can only be completed in class and presentations require your presence. If you are on the border of a higher grade, consistent attendance and active participation will work in your favor.

Laboratory Work

There will be 12 computer lab meetings. The labs are meant to teach you the basics of R and how to apply the material that is taught in class. In most laboratory sessions, there will be an assignment that should be completed during the session.

Note: Lab attendance is mandatory, and the TAs will take attendance. Unexcused absences will result in a zero on the corresponding lab assignment. Coordinate with your TA if an issue arises.

Midterm Exams

There will be two midterm exams, both in class. The first is focused more on coding and the second is more on theory. See the dates below.

Final Project and Presentation

Instead of a final exam, the course culminates in a final project and an in-class presentation. These are graded separately. The final project asks you to apply econometric methods from the course to a research question using real data. The presentation asks you to communicate your research question, data, empirical strategy, descriptive evidence, regression results, and limitations in a concise format. You should prepare slides for a 7-minute in-class presentation of your work. This will be followed by up to 3 minutes of Q&A. Project, proposal, and presentation guidelines, along with examples, are available on the course website. As a general rule, your analysis dataset should contain at least 1,000 observations. Projects using substantially smaller datasets require instructor approval.

Note: Attendance during all final project presentation sessions is mandatory, not only on the day of your own presentation. Listening to and engaging with classmates' presentations is an important component of the course. Any unexcused absence from a presentation session will result in a 20-point deduction from the final project grade for each session missed.

Authenticity of Work and Artificial Intelligence
AI tools may be used unless explicitly prohibited for a particular assignment. However, AI should be used as a complement to your learning, not as a substitute for understanding the material. You are responsible for understanding and being able to explain all submitted code, written responses, interpretations, and empirical choices.

If there is concern that submitted work does not reflect your own understanding, you may be asked to explain or defend your work orally. Work that cannot be adequately explained will receive a zero and may be referred under the university's academic integrity procedures.

Course Topics

1. Simple Linear Regression
2. Introduction to Causality
3. Multiple Linear Regression
4. Linear Regression Extensions
5. Sampling
6. Confidence Intervals and Hypothesis Testing
7. Regression Inference
8. Regression Discontinuity Designs
9. Difference-in-Differences
10. Panel Data

Important Dates

- October 9: One Page Proposal for Final Project Due
- October 15: Coding Midterm
- November 3: No Class (Election Day)
- November 5: Theory Midterm
- November 24, 26: No Class (Thanksgiving)
- Final Project Presentations: November 10, 12, 17, 19, December 1, 3
- Final Project: Due at the scheduled final exam time for this course. There will not be a final exam.

How to Be Successful in this Course
This is typically not considered to be an easy class. However, I have found that students that attend all classes and labs, complete all the material, read the material, are curious about the material, and put in a solid amount of work, all do very well.

Note: If you are doing all of these things and are still not doing well, please do not hesitate to contact me.

Your Well-being is Important
As a student you may experience a range of personal issues that can impede learning, such as strained relationships, increased anxiety, alcohol/drug concerns, feeling down, sadness, difficulty concentrating, lack of motivation, or other issues. These mental health concerns may impact your academic performance or your participation in daily activities. It is very important that you ask for help when you are struggling.

Time to Wait if Instructor is Delayed
If the instructor is late for class, please wait 15 minutes before leaving.

Information on Modality
All lectures and labs will be in-person and synchronous. Office hours will be conducted online, or in person by request.

Learning Environment
I will conduct the lectures (in person), but I encourage and look forward to student involvement in the lectures. This involvement will include questions, but it could also include students suggesting answers to the questions being posed, comments about the applicability of the material, discussions about the papers that we'll review, and so on.

Standard Academic Policies

Academic Integrity

As members of the Clemson University community, we have inherited Thomas Green Clemson's vision of this institution as a "high seminary of learning." Fundamental to this vision is a mutual commitment to truthfulness, honor, and responsibility, without which we cannot earn the trust and respect of others. Furthermore, we recognize that academic dishonesty detracts from the value of a Clemson degree. Therefore, we shall not tolerate lying, cheating, or stealing in any form.

All infractions of academic dishonesty by undergraduates must be reported to Undergraduate Learning for resolution through that office. In cases of plagiarism instructors may use the Plagiarism Resolution Form.

Additionally, for undergraduate classes:

Plagiarism, which includes the intentional or unintentional copying of language, structure, or ideas of another and attributing the work to one's own efforts. Graded works generated by artificial intelligence or ghostwritten (either paid or free) are expressly forbidden.
See the Undergraduate Academic Integrity Policy website for additional information and the current catalog ("Academic Regulations" section) for the policy. Send questions to UGSintegrity@clemson.edu.

Accessibility

Students with disabilities or temporary injuries/conditions may require accommodations due to barriers in the structure of facilities, course design, technology used for curricular purposes, or other campus resources. Students who experience a barrier to full access to this class should let the instructor know and are encouraged to request accommodations through SAS (Student Accessibility Services) as soon as possible. To request accommodations through SAS, please see this link: www.clemson.edu/academics/student-accessibility-services/how-to-register/requesting-accommodations. You can also reach out to SAS with questions by calling 864-656-6848, email CUSAS@clemson.edu or visiting SAS at the ASC Suite 239. Contact the office for the most updated drop-in schedule if you would prefer not to schedule an appointment.

The Clemson University Title IX Statement Regarding Non-Discrimination
Clemson University is committed to a policy of equal opportunity for all persons and does not discriminate on the basis of race, color, religion, sex, sexual orientation, gender, pregnancy or related conditions (including pregnancy, childbirth, termination of pregnancy, lactation, recovery from the foregoing, or medical conditions related to the foregoing), national origin, age, disability, veteran's status, genetic information or protected activity in employment, educational programs and activities, admissions and financial aid. This includes a prohibition against sex discrimination (including sex-based harassment and sexual violence) as mandated by Title IX of the Education Amendments of 1972. This Title IX policy is located on the Access Compliance and Education website. Ms. Alesia Smith is the Clemson University Title IX Coordinator, and the Assistant Vice President of Equity Compliance. Her office is located at 223 Brackett Hall, 864-656-3181 and her email address is alesias@clemson.edu. Remember, email is not a fully secured method of communication and should not be used to discuss Title IX issues.
Clemson University aspires to create a diverse community that welcomes people of different races, cultures, ages, genders, sexual orientation, religions, socioeconomic levels, political perspectives, abilities, opinions, values and experiences.

Emergency Preparation

Emergency procedures have been posted in all buildings and on all elevators. Students should be reminded to review these procedures for their own safety. All students and employees should be familiar with guidelines from Clemson University Public Safety.
Clemson University is committed to providing a safe campus environment for students, faculty, staff, and visitors. As members of the community, we encourage you to take the following actions to be better prepared in case of an emergency:

1. Familiarize yourself with all possible exits, safer locations, and other key information on the emergency evacuation maps in this building, and those that you visit regularly.
2. Make a plan for how you would Run, Hide, and Fight in case of an active threat in this building, and those that you visit regularly. For example:
    a. Run – what are all the possible exits in this building, and the routes to them?
    b. Hide – what are the potential hiding locations in this room and building that are out of sight of doors and windows, how do you lock the door(s), how would you barricade the door(s) and windows, where do you turn off the lights?
    c. Fight – What tools are available in this room and building, should you have to fight?
3. Ensure you are signed up for emergency alerts. Alerts are only sent when there is a potential threat to safety, a major disruption to campus services, and once-monthly tests.
4. Download the Rave Guardian app to your phone. (https://www.clemson.edu/cusafety/cupd/rave-guardian/)
5. Learn what you can do to prepare yourself for the hazards that affect our locations. (http://www.clemson.edu/cusafety/EmergencyManagement/)